The Journal of Open Source Software

SegregatedVMSSolver.jl: Linearized and Segregated Stabilized Solver for Large Eddy Simulation in Julia

Carlo Brunelli [1]

1 Mechanical Engineering Department, Royal Military Academy, Belgium

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Kevin M. Moerman

Reviewers:

- @marinlauber
- @nathanquinlan
- @weymouth

Submitted: 19 May 2024
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

Large-Eddy Simulation (LES) is a family of mathematical techniques that perform high-fidelity simulations in Computational Fluid Dynamics (CFD). They can simulate turbulent flows by numerically solving the Navier-Stokes equations. Using filtering operation, LES focuses on the larger length scale, while the effects of small scales (subgrid scales) are modeled. The Variational MultiScale (VMS) and Streamline-Upwind/Petrov-Galerkin (SUPG) methods belong to the family of stabilized methods. The mathematical problem is solved using the Finite Element Method (FEM) framework. They allow to simulate complex flows. The package has been developed with the primary aim of studying the Laminar Separation Bubble (LSB) at low-Reynolds regime on the suction side of the airfoils.

One of the primary objectives of `SegregatedVMSSolver.jl` is to empower users with different levels of expertise in CFD to run simulations seamlessly. For users with a basic understanding of CFD principles, the package aims to provide a framework where simulations can be executed without the need for detailed parameters input.

However, for more advanced users who may desire greater control over the simulation process, it is possible to provide custom options for customizing almost all aspects of the method: solver, model, exports and HPC options. This customization empowers users to fine-tune parameters according to their specific requirements.

Balancing between simplicity for novice users and flexibility for advanced users, the package ensures usability without compromising on the depth and sophistication of the simulation capabilities. This approach makes the package versatile and adaptable to a wide range of users, from researchers seeking quick insights to seasoned practitioners, using HPC, aiming for comprehensive analyses in fluid dynamics applications.

Statement of need

`SegregatedVMSSolver.jl` is a comprehensive Julia package designed for conducting high-fidelity simulations of complex flow phenomena whithin the incompressible regime, leveraging VMS and SUPG method. VMS has been originally introduced by (Hughes et al., 2000). The linearization adopted has been proposed in the SUPG method by (Banyai, 2016).

The package relies on `Gridap.jl`(Verdugo & Badia, 2022),(Badia & Verdugo, 2020) package to implement the mathematical model of FEM. Complementing this core functionality, `GridapDistributed.jl`(Badia et al., 2022) and `PartitionedArrays.jl` allow to use multi-core CPU desktop computers to HPC clusters. The `GridapPETSc.jl` package is used to solve the final linear system.

It solves a Linearized and Segregated version of VMS (LS-VMS) and SUPG. It is based on an iterative predictor-corrector scheme and it avoids the resolution of a non-linear problem. It

uses the θ method to solve the time-marching problem.

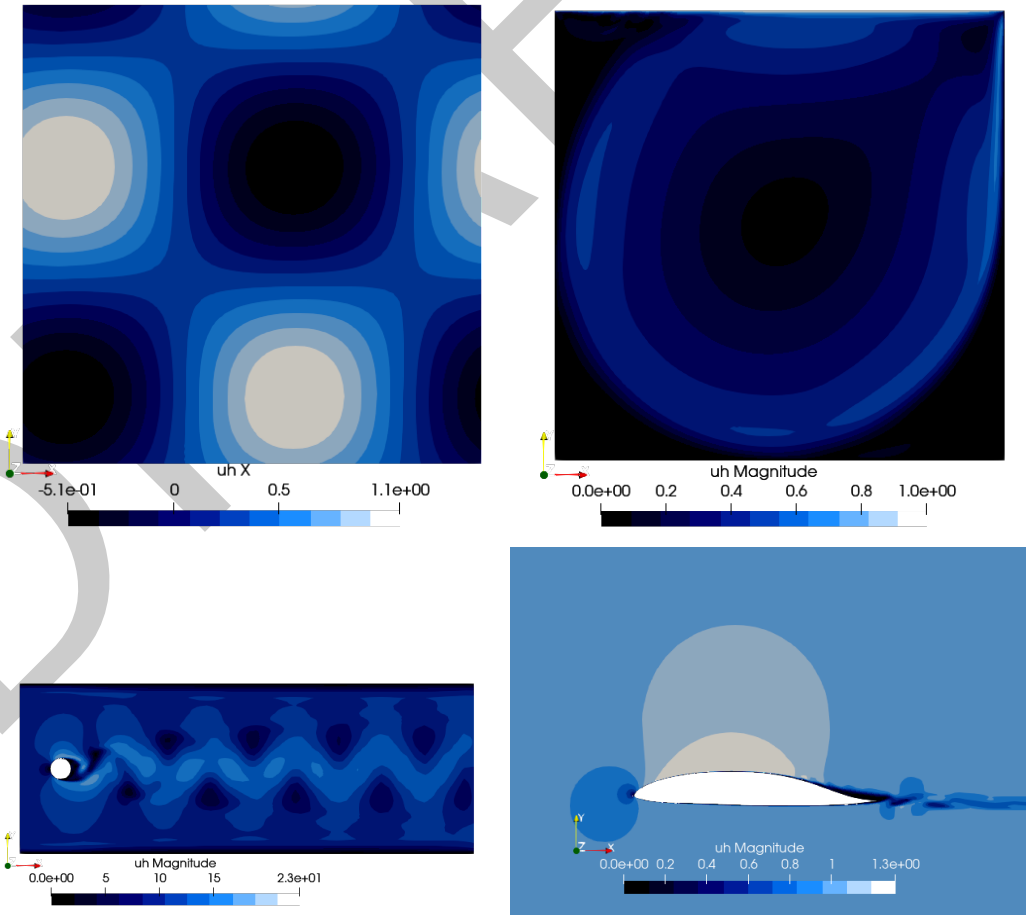
A suite of different problems are encoded: Taylor-Green vortices (2D), lid-driven cavity, vortex shedding over a cylinder and airfoil. The package is actually dedicated to study airfoils aerodynamic features low-Reynolds regime (up to Renynolds number 500000).

Different 3D problems, including Taylor-Green vortices (only 2D), lid-driven cavity, vortex shedding over a cylinder, and airfoil aerodynamics within the low-Reynolds regime (tested up to Reynolds number $500,000$), are meticulously coded for comprehensive analysis.

It has a suite of tools for post-processing the results like performing time and spanwise averaging, obtaining. It is possible to control the simulation in real-time, enabling/disabling the creation of output files at each time-step. The user can load own airfoil meshes also usign the package `AirfoilGmsh.jl`. Additionally, the package provides tools for initializing boundary layers in 2D simulations, enhancing its versatility and utility for researchers and practitioners in the field of fluid dynamics.

Different software packages have been developed in the field of fluid dynamics in Julia: Julia Packages fluid-dynamics showing a growing interest in the field by the Julia community.

Results



DRAFT

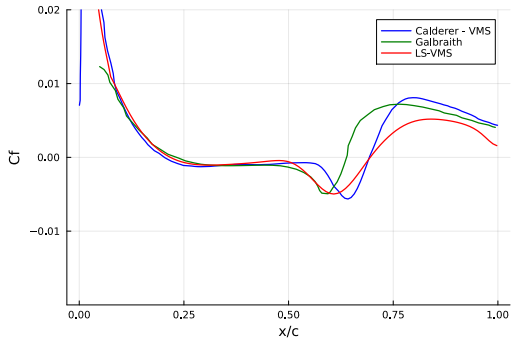


Figure 1: Friction coefficient on the suction side of the sd7003 airfoil, Reynolds 60 000, angle of attack 4°,

Figure 1 shows the comparison of the time averaged results obtained using the LS-VMS, compared with the results obtained by Calderer et al. (Calderer & Masud, 2013) and Galbraith (Galbraith et al., 2008).

Package Features

- Support 2-dimensional and 3-dimensional geometries
- It solved a time-dependent problem
- It can run in parallel using MPI
- Velocity ramping
- It can capture the laminar-to-turbulent transition
- It has utilities for time and space averaging
- It supports advanced boundary layer initialization
- Real-time simulation interaction
- The solution can be visualized in ParaView
- Restart the simulation from a specific saved time-step

Acknowledgements

The authors would like to acknowledge the Royal Higher Institute for Defense for funding this research through the project MSP21/02 Numerical and Experimental Low Speed High Altitude Wing (NELSHAW). Many thanks professor Bart Janssens for his invaluable support, patience and feedback.

References

Badia, S., Martín, A. F., & Verdugo, F. (2022). GridapDistributed: A massively parallel finite element toolbox in julia. *Journal of Open Source Software*, *7*(74), 4157. https://doi.org/10.21105/joss.04157

Badia, S., & Verdugo, F. (2020). Gridap: An extensible finite element toolbox in julia. *Journal of Open Source Software*, *5*(52), 2520. https://doi.org/10.21105/joss.02520

Banyai, T. (2016). *Development of stabilized finite element method for numerical simulation of turbulent incompressible single and eulerian-eulerian two-phase flows*. ISBN: 9782875161093

Calderer, R., & Masud, A. (2013). Residual-based variational multiscale turbulence models for unstructured tetrahedral meshes. *Computer Methods in Applied Mechanics and Engineering*, *254*, 238–253. https://doi.org/10.1016/j.cma.2012.09.015

DRAFT

Galbraith, M. C., Visbal, M. R., & Student, G. (2008). *Implicit large eddy simulation of low reynolds number flow past the SD7003 airfoil*.

Hughes, T. J. R., Mazzei, L., & Jansen, K. E. (2000). Computing and visualization in science large eddy simulation and the variational multiscale method. In *Comput Visual Sci* (Vol. 3, pp. 47–59).

Verdugo, F., & Badia, S. (2022). The software design of gridap: A finite element package based on the julia JIT compiler. *Computer Physics Communications*, *276*, 108341. https://doi.org/10.1016/j.cpc.2022.108341

DRAFT